

Manual J Report



Created by	nicholas.wood@energyefficiencyservices.org
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Manual J Version	8th Edition

Name	1 Story Handicap End Unit With Basement
Address	2506 Frazier Ave, Madison, WI 53713, USA

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Design Conditions

• ASHRAE Weather Station	DANE COUNTY REGIONAL AP, WI
• Elevation	878ft
• Latitude	43.047
• Altitude Correction Factor	0.9737

	Outdoor Dry-bulb	Indoor Dry-bulb	Temperature Difference	Indoor RH	Daily Range	Grains Difference
Heating	-1F	70F	71F	N/A	N/A	N/A
Cooling	87F	75F	12F	50%	Medium	32

WHOLE HOME

Load Summary

• Total Floor Area (ft ²)	1,881
• Total Volume (ft ³)	14,853
• Ceiling height(s) (ft)	8
• Running length of exposed wall (ft)	118
• Total glazing area (ft ²)	69
• Sensible Heat Ratio (SHR)	0.872
• Exposed wall area (ft ²) (gross) - Total	1,670
◦ N	477
◦ E	467
◦ W	259
◦ S	467
• Exposed wall area (ft ²) (net)	1,636
• Envelope tightness	Semi-Loose
• Appliance Scenario	Default Appliance Load
• Number of occupants	3

	Sensible Cooling BTUH	Latent Cooling BTUH	Total Cooling BTUH	Total Heating BTUH
Total	10,468	1,534	12,002	46,735
Walls	1,239	0	1,239	15,796
Glazing	2,470	0	2,470	2,792
Doors	0	0	0	0
Ceilings	2,092	0	2,092	6,532
Floors	471	0	471	8,338
Ventilation	0	0	0	0
Ducts	862	1,004	1,866	5,233
Infiltration Loss and Gain	198	331	529	8,044
Blower Heat	1,706	0	1,706	0
Occupants	230	200	430	0
Appliances	1,200	0	1,200	0
Plants	0	0	0	0
Hot Water Piping	0	0	0	0
Moisture Migration	0	0	0	0
Winter Humidification	0	0	0	0

GAS FURNACE

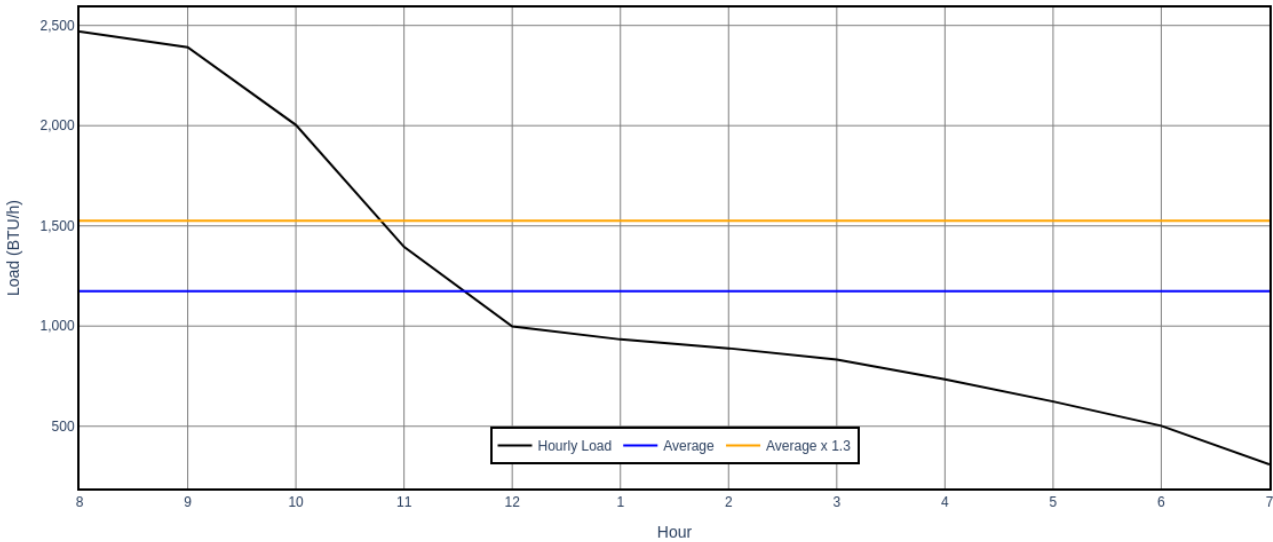
Load Summary

• Total Floor Area (ft ²)	908
• System Type	Heating
• Distribution Type	Ducted
• Ducts	Ducts 1
◦ Supply Run Location	Unconditioned basement or closed crawl space
◦ Leakage Class	Poor (24 CFM/100 sq ft of supply duct surface area, 47 CFM/100 sq ft for returns)
◦ Duct Wall Insulation	R-4 (typical of Flexduct)
◦ Airway Configuration	Trunk and Branch

◦ EHLF	0.18
◦ ESGF	0.11
◦ ELG	1004

	Sensible Cooling BTUH	Latent Cooling BTUH	Total Cooling BTUH	Total Heating BTUH
Total	0	0	0	16,853
Walls	0	0	0	4,170
Glazing	0	0	0	1,396
Doors	0	0	0	0
Ceilings	0	0	0	3,160
Floors	0	0	0	3,282
Ventilation	0	0	0	0
Ducts	0	0	0	2,617
Infiltration Loss and Gain	0	0	0	2,229
Blower Heat	0	0	0	0
Occupants	0	0	0	0
Appliances	0	0	0	0
Plants	0	0	0	0
Hot Water Piping	0	0	0	0
Moisture Migration	0	0	0	0
Winter Humidification	0	0	0	0

Gas Furnace - Hourly Fenestration Load - Midsummer

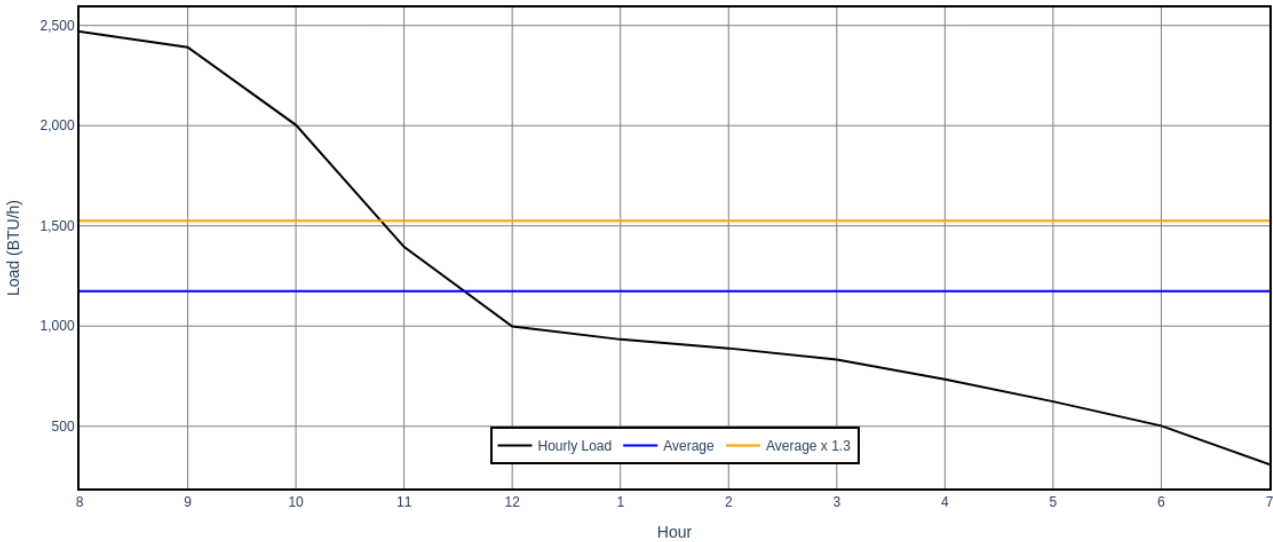


Zone 1 Load Summary

• Total Floor Area (ft²)	908
• Rooms	First story - Kitchen (235 ft ²)
	First story - Bedroom 2 (169 ft ²)
	First story - Bathroom (68 ft ²)
	First story - Bedroom 1 (165 ft ²)
	First story - Living Room (271 ft ²)

	Sensible Cooling BTUH	Latent Cooling BTUH	Total Cooling BTUH	Total Heating BTUH
Total	0	0	0	16,853
Walls	0	0	0	4,170
Glazing	0	0	0	1,396
Doors	0	0	0	0
Ceilings	0	0	0	3,160
Floors	0	0	0	3,282
Ventilation	0	0	0	0
Ducts	0	0	0	2,617
Infiltration Loss and Gain	0	0	0	2,229
Blower Heat	0	0	0	0
Occupants	0	0	0	0
Appliances	0	0	0	0
Plants	0	0	0	0
Hot Water Piping	0	0	0	0
Moisture Migration	0	0	0	0
Winter Humidification	0	0	0	0

Zone 1 - Hourly Fenestration Load - Midsummer



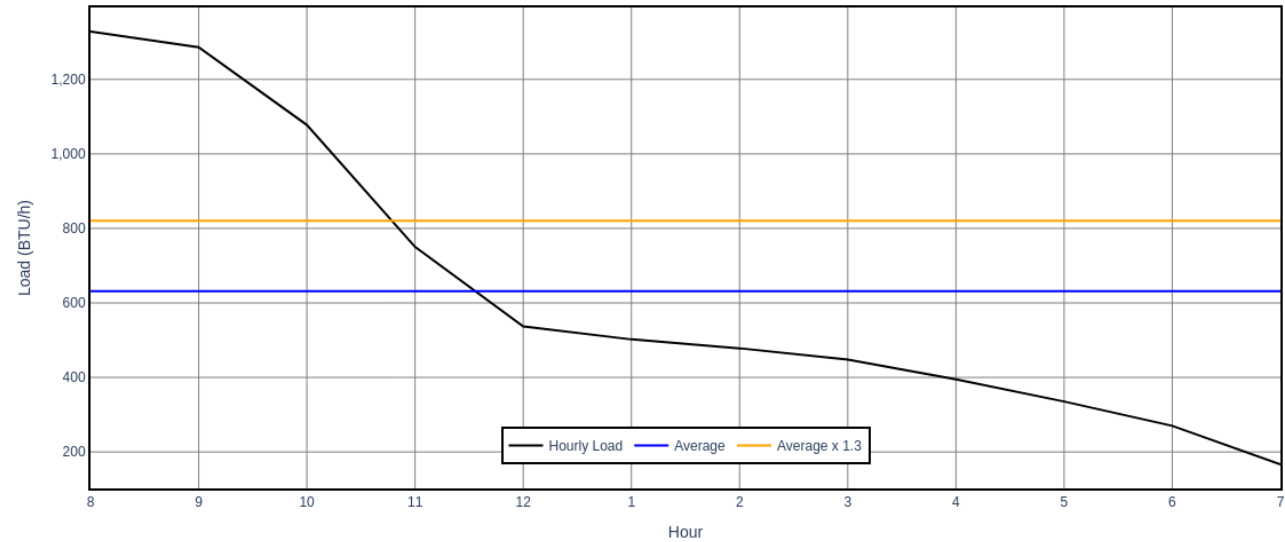
Kitchen Load

First story

• Total Floor Area (ft ²)	235
• Total Volume (ft ³)	1,886
• Ceiling height(s) (ft)	8.0
• Exposed wall area (ft ²) (gross)	226
• Exposed wall area (ft ²) (net)	207
• Total glazing area (ft ²)	19
• Design CFM	
• Occupants	0

	Sensible Cooling BTUH	Latent Cooling BTUH	Total Cooling BTUH	Total Heating BTUH
Total	0	0	0	5,484
Walls	0	0	0	1,428
Glazing	0	0	0	751
Doors	0	0	0	0
Ceilings	0	0	0	818
Floors	0	0	0	849
Ventilation	0	0	0	0
Ducts	0	0	0	852
Infiltration Loss and Gain	0	0	0	787
Blower Heat	0	0	0	0
Occupants	0	0	0	0
Appliances	0	0	0	0
Plants	0	0	0	0
Hot Water Piping	0	0	0	0
Moisture Migration	0	0	0	0
Winter Humidification	0	0	0	0

Kitchen (First) - Hourly Fenestration Load - Midsummer



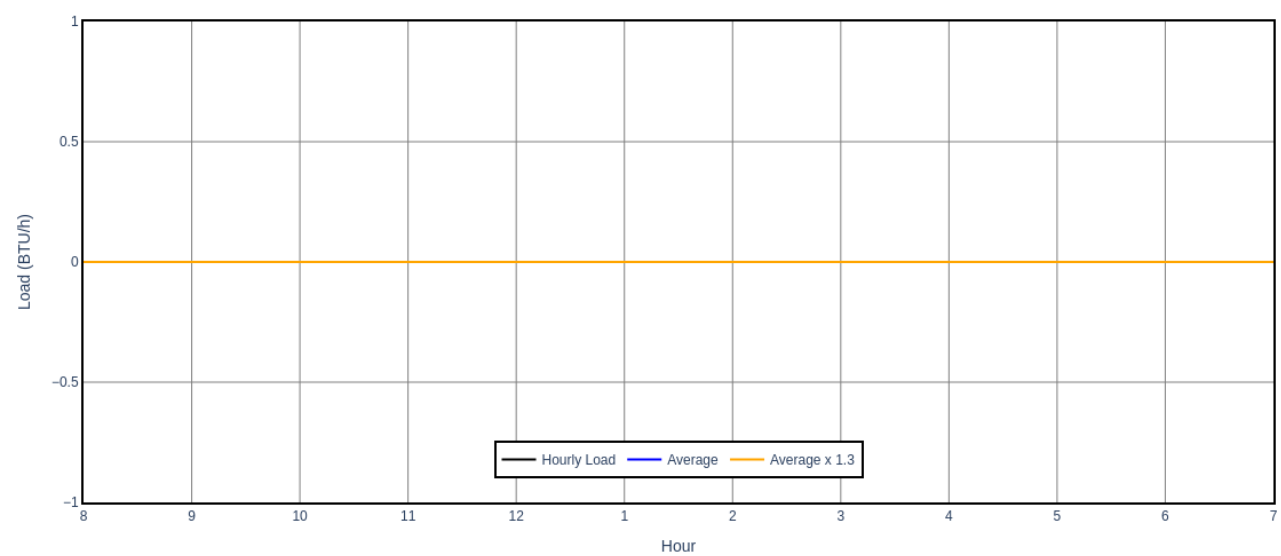
Bedroom 2 Load

First story

• Total Floor Area (ft ²)	169
• Total Volume (ft ³)	1,356
• Ceiling height(s) (ft)	8.0
• Exposed wall area (ft ²) (gross)	102
• Exposed wall area (ft ²) (net)	102
• Total glazing area (ft ²)	0
• Design CFM	
• Occupants	0

	Sensible Cooling BTUH	Latent Cooling BTUH	Total Cooling BTUH	Total Heating BTUH
Total	0	0	0	2,676
Walls	0	0	0	706
Glazing	0	0	0	0
Doors	0	0	0	0
Ceilings	0	0	0	588
Floors	0	0	0	610
Ventilation	0	0	0	0
Ducts	0	0	0	416
Infiltration Loss and Gain	0	0	0	357
Blower Heat	0	0	0	0
Occupants	0	0	0	0
Appliances	0	0	0	0
Plants	0	0	0	0
Hot Water Piping	0	0	0	0
Moisture Migration	0	0	0	0
Winter Humidification	0	0	0	0

Bedroom 2 (First) - Hourly Fenestration Load - Midsummer



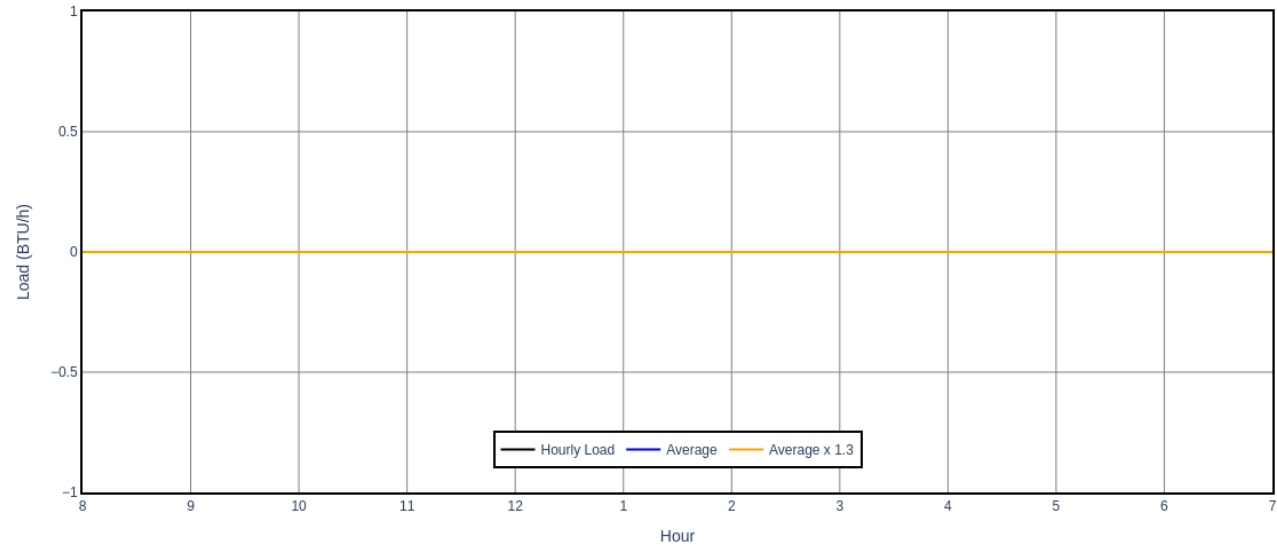
Bathroom Load

First story

• Total Floor Area (ft ²)	68
• Total Volume (ft ³)	544
• Ceiling height(s) (ft)	8.0
• Exposed wall area (ft ²) (gross)	39
• Exposed wall area (ft ²) (net)	39
• Total glazing area (ft ²)	0
• Design CFM	
• Occupants	0

	Sensible Cooling BTUH	Latent Cooling BTUH	Total Cooling BTUH	Total Heating BTUH
Total	0	0	0	1,046
Walls	0	0	0	268
Glazing	0	0	0	0
Doors	0	0	0	0
Ceilings	0	0	0	236
Floors	0	0	0	245
Ventilation	0	0	0	0
Ducts	0	0	0	162
Infiltration Loss and Gain	0	0	0	135
Blower Heat	0	0	0	0
Occupants	0	0	0	0
Appliances	0	0	0	0
Plants	0	0	0	0
Hot Water Piping	0	0	0	0
Moisture Migration	0	0	0	0
Winter Humidification	0	0	0	0

Bathroom (First) - Hourly Fenestration Load - Midsummer



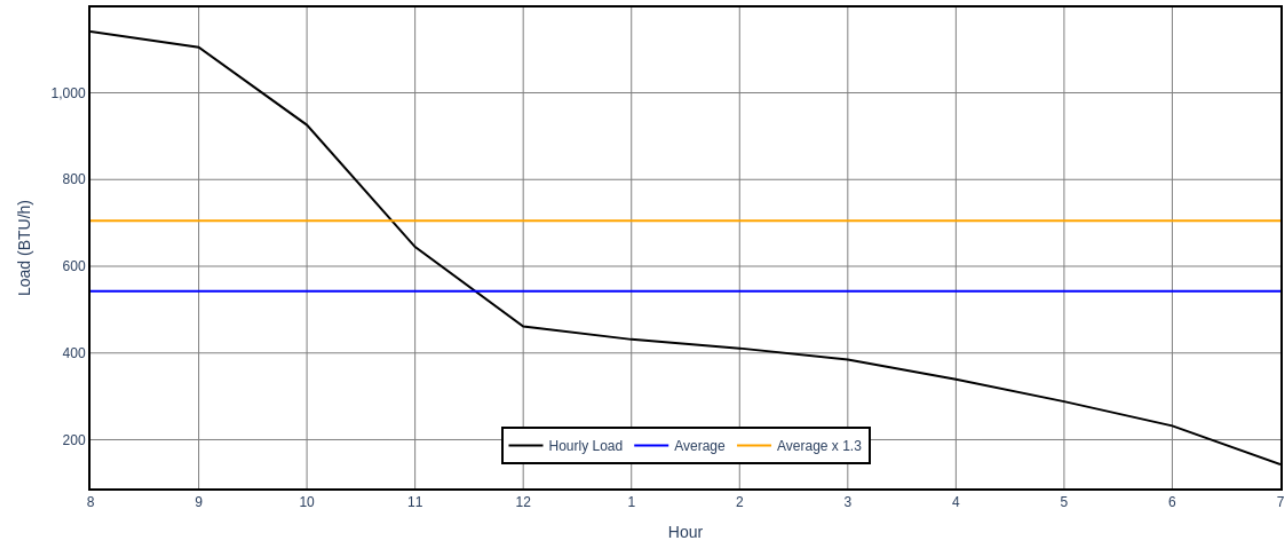
Bedroom 1 Load

First story

• Total Floor Area (ft ²)	165
• Total Volume (ft ³)	1,325
• Ceiling height(s) (ft)	8.0
• Exposed wall area (ft ²) (gross)	210
• Exposed wall area (ft ²) (net)	194
• Total glazing area (ft ²)	16
• Design CFM	
• Occupants	0

	Sensible Cooling BTUH	Latent Cooling BTUH	Total Cooling BTUH	Total Heating BTUH
Total	0	0	0	4,601
Walls	0	0	0	1,338
Glazing	0	0	0	645
Doors	0	0	0	0
Ceilings	0	0	0	574
Floors	0	0	0	597
Ventilation	0	0	0	0
Ducts	0	0	0	714
Infiltration Loss and Gain	0	0	0	732
Blower Heat	0	0	0	0
Occupants	0	0	0	0
Appliances	0	0	0	0
Plants	0	0	0	0
Hot Water Piping	0	0	0	0
Moisture Migration	0	0	0	0
Winter Humidification	0	0	0	0

Bedroom 1 (First) - Hourly Fenestration Load - Midsummer



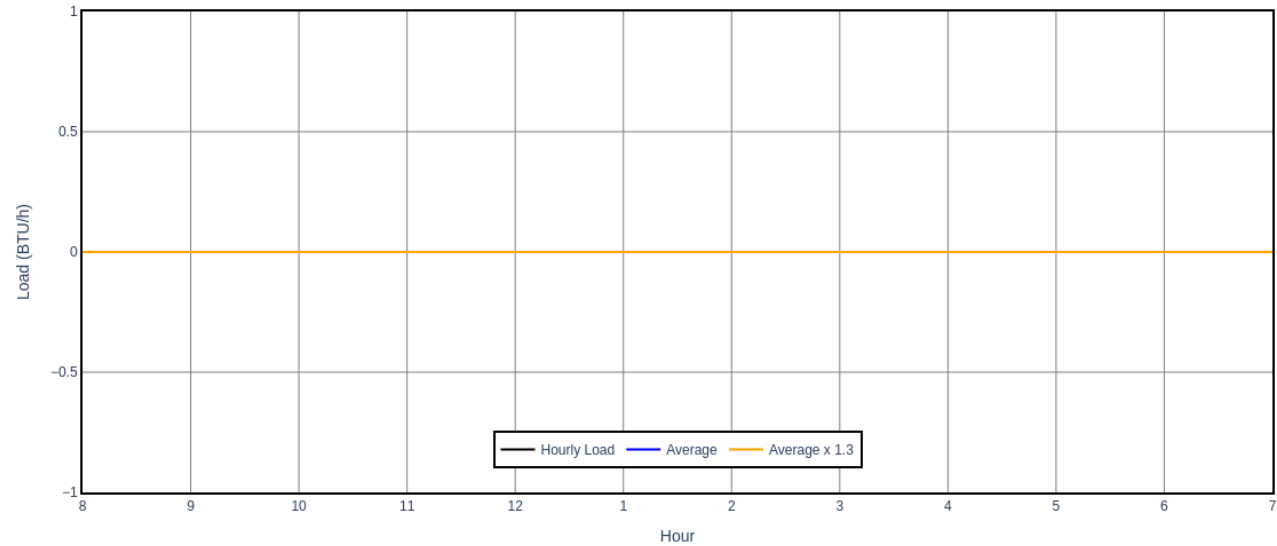
Living Room Load

First story

• Total Floor Area (ft ²)	271
• Total Volume (ft ³)	2,177
• Ceiling height(s) (ft)	8.0
• Exposed wall area (ft ²) (gross)	63
• Exposed wall area (ft ²) (net)	63
• Total glazing area (ft ²)	0
• Design CFM	
• Occupants	1

	Sensible Cooling BTUH	Latent Cooling BTUH	Total Cooling BTUH	Total Heating BTUH
Total	0	0	0	3,046
Walls	0	0	0	431
Glazing	0	0	0	0
Doors	0	0	0	0
Ceilings	0	0	0	944
Floors	0	0	0	980
Ventilation	0	0	0	0
Ducts	0	0	0	473
Infiltration Loss and Gain	0	0	0	218
Blower Heat	0	0	0	0
Occupants	0	0	0	0
Appliances	0	0	0	0
Plants	0	0	0	0
Hot Water Piping	0	0	0	0
Moisture Migration	0	0	0	0
Winter Humidification	0	0	0	0

Living Room (First) - Hourly Fenestration Load - Midsummer

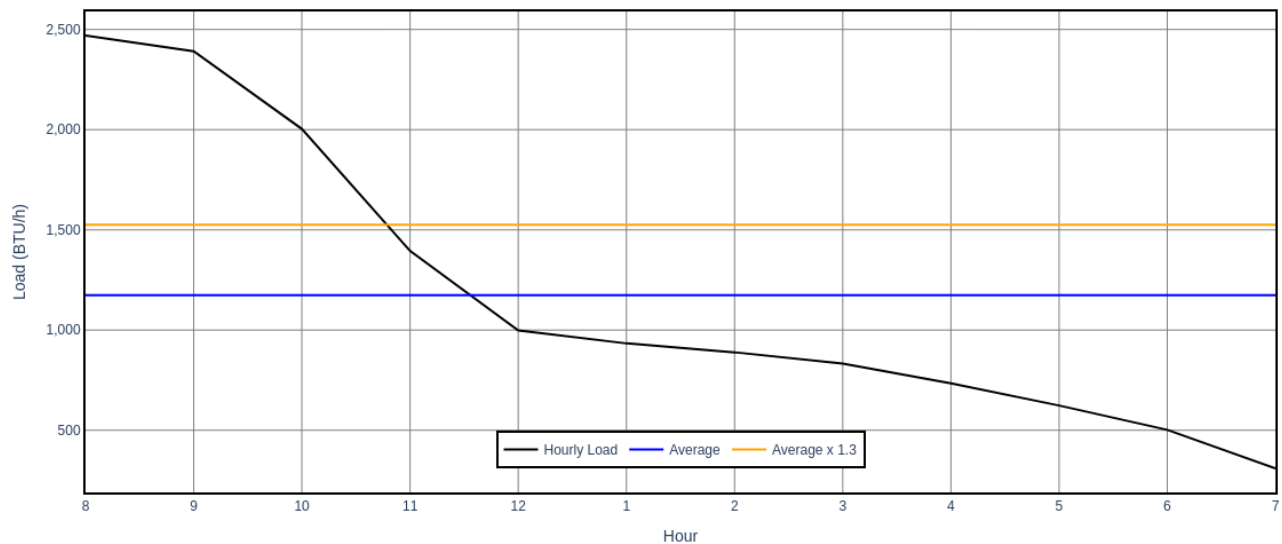


Load Summary

- **Total Floor Area (ft²)** 908
- **System Type** Heating and Cooling
- **Distribution Type** Ducted
- **Ducts** Ducts 1
 - **Supply Run Location** Unconditioned basement or closed crawl space
 - **Leakage Class** Poor (24 CFM/100 sq ft of supply duct surface area, 47 CFM/100 sq ft for returns)
 - **Duct Wall Insulation** R-4 (typical of Flexduct)
 - **Airway Configuration** Trunk and Branch
 - **EHLF** 0.18
 - **ESGF** 0.11
 - **ELG** 1004

	Sensible Cooling BTUH	Latent Cooling BTUH	Total Cooling BTUH	Total Heating BTUH
Total	10,468	1,534	12,002	16,853
Walls	1,239	0	1,239	4,170
Glazing	2,470	0	2,470	1,396
Doors	0	0	0	0
Ceilings	2,092	0	2,092	3,160
Floors	471	0	471	3,282
Ventilation	0	0	0	0
Ducts	862	1,004	1,866	2,617
Infiltration Loss and Gain	198	331	529	2,229
Blower Heat	1,706	0	1,706	0
Occupants	230	200	430	0
Appliances	1,200	0	1,200	0
Plants	0	0	0	0
Hot Water Piping	0	0	0	0
Moisture Migration	0	0	0	0
Winter Humidification	0	0	0	0

Cold Climate Heat Pump - Hourly Fenestration Load - Midsummer

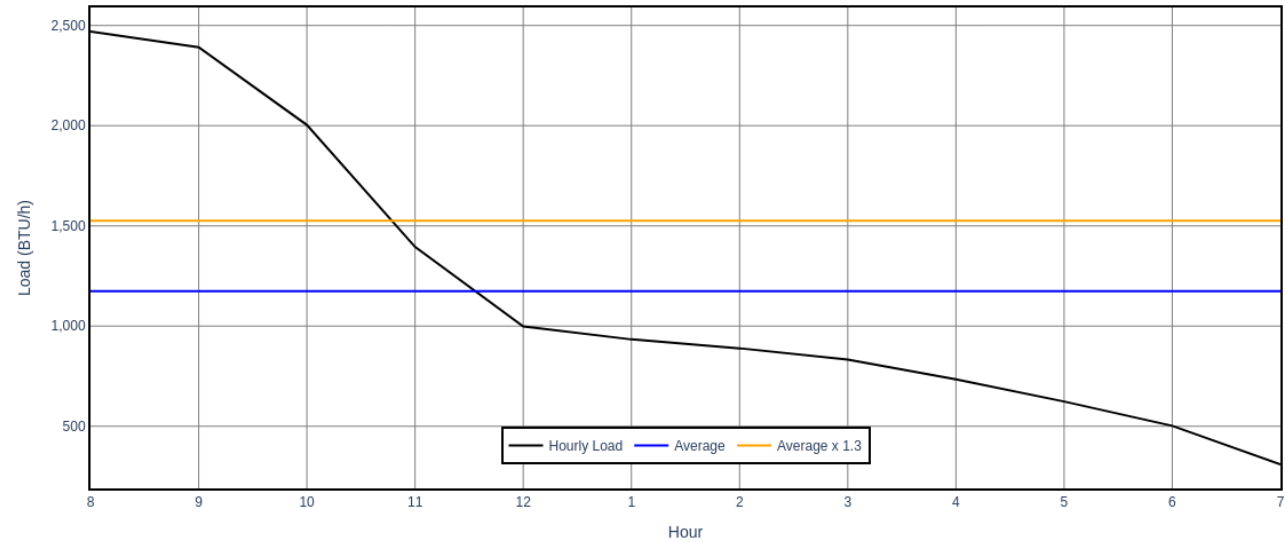


Zone 1 Load Summary

• Total Floor Area (ft ²)	908
• Rooms	First story - Bedroom 2 (169 ft ²)
	First story - Bathroom (68 ft ²)
	First story - Bedroom 1 (165 ft ²)
	First story - Living Room (271 ft ²)
	First story - Kitchen (235 ft ²)

	Sensible Cooling BTUH	Latent Cooling BTUH	Total Cooling BTUH	Total Heating BTUH
Total	8,762	1,534	10,296	16,853
Walls	1,239	0	1,239	4,170
Glazing	2,470	0	2,470	1,396
Doors	0	0	0	0
Ceilings	2,092	0	2,092	3,160
Floors	471	0	471	3,282
Ventilation	0	0	0	0
Ducts	862	1,004	1,866	2,617
Infiltration Loss and Gain	198	331	529	2,229
Blower Heat	0	0	0	0
Occupants	230	200	430	0
Appliances	1,200	0	1,200	0
Plants	0	0	0	0
Hot Water Piping	0	0	0	0
Moisture Migration	0	0	0	0
Winter Humidification	0	0	0	0

Zone 1 - Hourly Fenestration Load - Midsummer



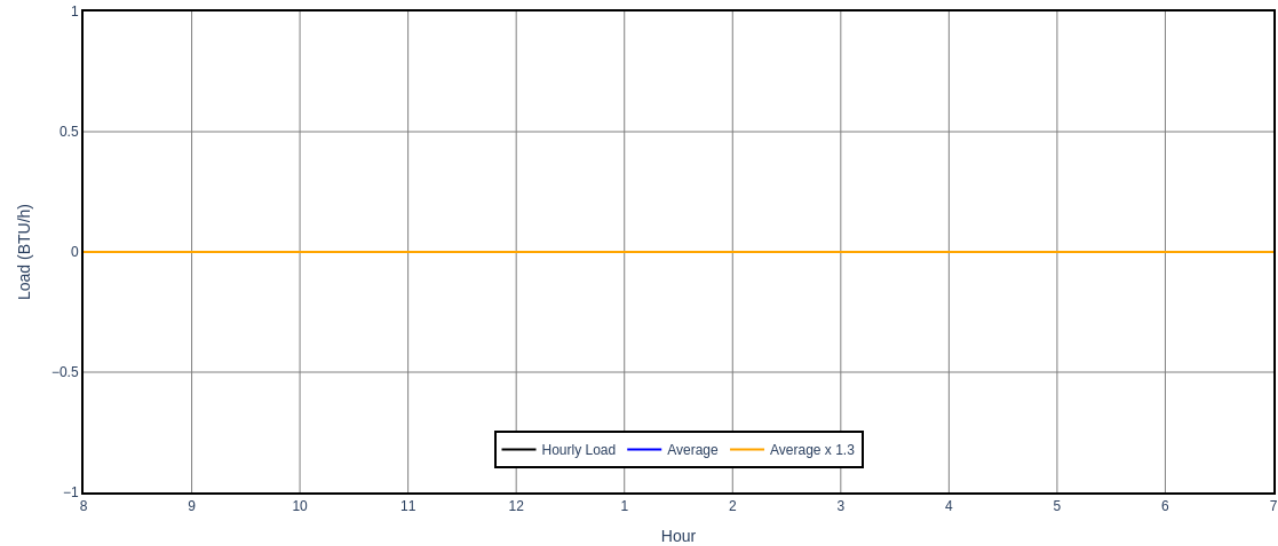
Bedroom 2 Load

First story

• Total Floor Area (ft²)	169
• Total Volume (ft³)	1,356
• Ceiling height(s) (ft)	8.0
• Exposed wall area (ft²) (gross)	102
• Exposed wall area (ft²) (net)	102
• Total glazing area (ft²)	0
• Design CFM	
• Occupants	0

	Sensible Cooling BTUH	Latent Cooling BTUH	Total Cooling BTUH	Total Heating BTUH
Total	796	144	941	2,676
Walls	210	0	210	706
Glazing	0	0	0	0
Doors	0	0	0	0
Ceilings	389	0	389	588
Floors	88	0	88	610
Ventilation	0	0	0	0
Ducts	78	91	170	416
Infiltration Loss and Gain	32	53	85	357
Blower Heat	0	0	0	0
Occupants	0	0	0	0
Appliances	0	0	0	0
Plants	0	0	0	0
Hot Water Piping	0	0	0	0
Moisture Migration	0	0	0	0
Winter Humidification	0	0	0	0

Bedroom 2 (First) - Hourly Fenestration Load - Midsummer



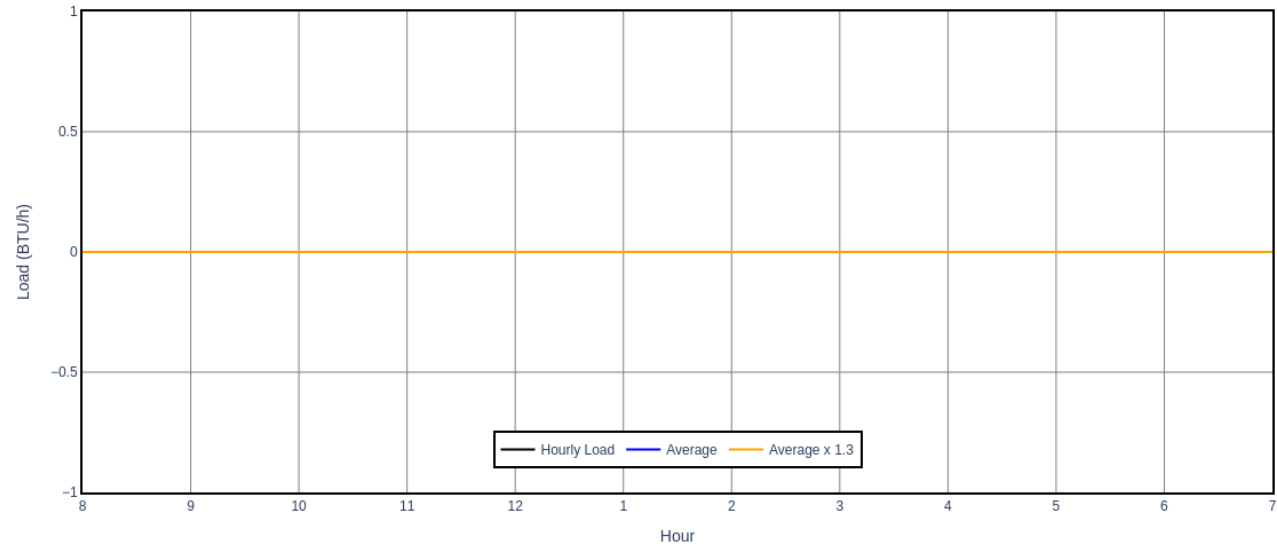
Bathroom Load

First story

• Total Floor Area (ft ²)	68
• Total Volume (ft ³)	544
• Ceiling height(s) (ft)	8.0
• Exposed wall area (ft ²) (gross)	39
• Exposed wall area (ft ²) (net)	39
• Total glazing area (ft ²)	0
• Design CFM	
• Occupants	0

	Sensible Cooling BTUH	Latent Cooling BTUH	Total Cooling BTUH	Total Heating BTUH
Total	314	56	370	1,046
Walls	80	0	80	268
Glazing	0	0	0	0
Doors	0	0	0	0
Ceilings	156	0	156	236
Floors	35	0	35	245
Ventilation	0	0	0	0
Ducts	31	36	67	162
Infiltration Loss and Gain	12	20	32	135
Blower Heat	0	0	0	0
Occupants	0	0	0	0
Appliances	0	0	0	0
Plants	0	0	0	0
Hot Water Piping	0	0	0	0
Moisture Migration	0	0	0	0
Winter Humidification	0	0	0	0

Bathroom (First) - Hourly Fenestration Load - Midsummer



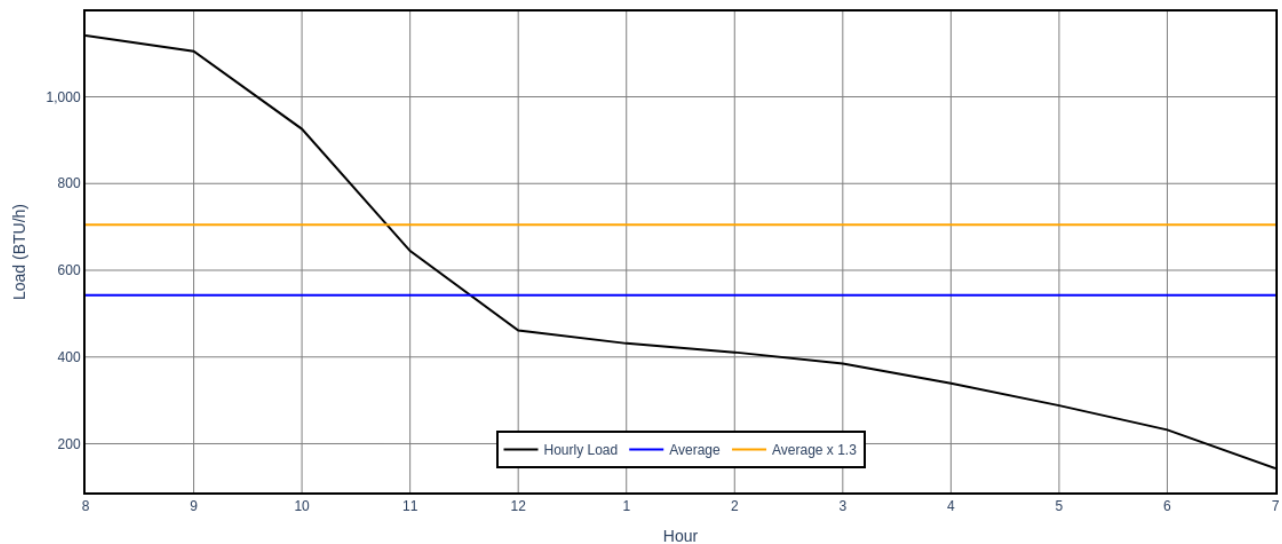
Bedroom 1 Load

First story

• Total Floor Area (ft ²)	165
• Total Volume (ft ³)	1,325
• Ceiling height(s) (ft)	8.0
• Exposed wall area (ft ²) (gross)	210
• Exposed wall area (ft ²) (net)	194
• Total glazing area (ft ²)	16
• Design CFM	
• Occupants	0

	Sensible Cooling BTUH	Latent Cooling BTUH	Total Cooling BTUH	Total Heating BTUH
Total	2,296	372	2,667	4,601
Walls	398	0	398	1,338
Glazing	1,141	0	1,141	645
Doors	0	0	0	0
Ceilings	380	0	380	574
Floors	86	0	86	597
Ventilation	0	0	0	0
Ducts	226	263	489	714
Infiltration Loss and Gain	65	109	174	732
Blower Heat	0	0	0	0
Occupants	0	0	0	0
Appliances	0	0	0	0
Plants	0	0	0	0
Hot Water Piping	0	0	0	0
Moisture Migration	0	0	0	0
Winter Humidification	0	0	0	0

Bedroom 1 (First) - Hourly Fenestration Load - Midsummer



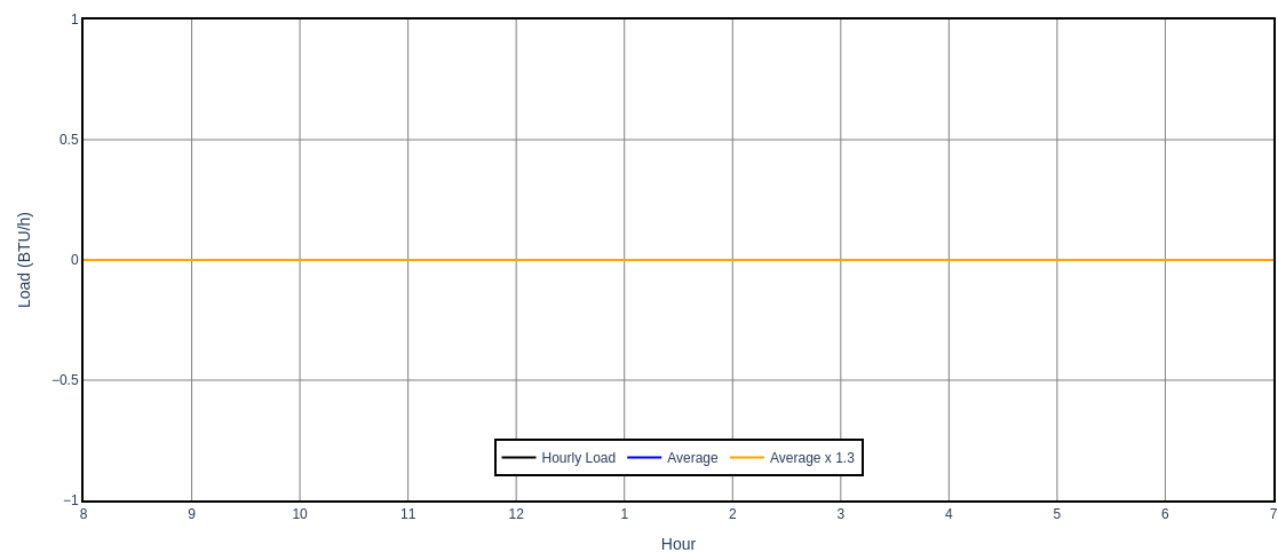
Living Room Load

First story

• Total Floor Area (ft ²)	271
• Total Volume (ft ³)	2,177
• Ceiling height(s) (ft)	8.0
• Exposed wall area (ft ²) (gross)	63
• Exposed wall area (ft ²) (net)	63
• Total glazing area (ft ²)	0
• Design CFM	
• Occupants	1

	Sensible Cooling BTUH	Latent Cooling BTUH	Total Cooling BTUH	Total Heating BTUH
Total	1,268	378	1,645	3,046
Walls	128	0	128	431
Glazing	0	0	0	0
Doors	0	0	0	0
Ceilings	625	0	625	944
Floors	141	0	141	980
Ventilation	0	0	0	0
Ducts	125	145	270	473
Infiltration Loss and Gain	19	32	52	218
Blower Heat	0	0	0	0
Occupants	230	200	430	0
Appliances	0	0	0	0
Plants	0	0	0	0
Hot Water Piping	0	0	0	0
Moisture Migration	0	0	0	0
Winter Humidification	0	0	0	0

Living Room (First) - Hourly Fenestration Load - Midsummer



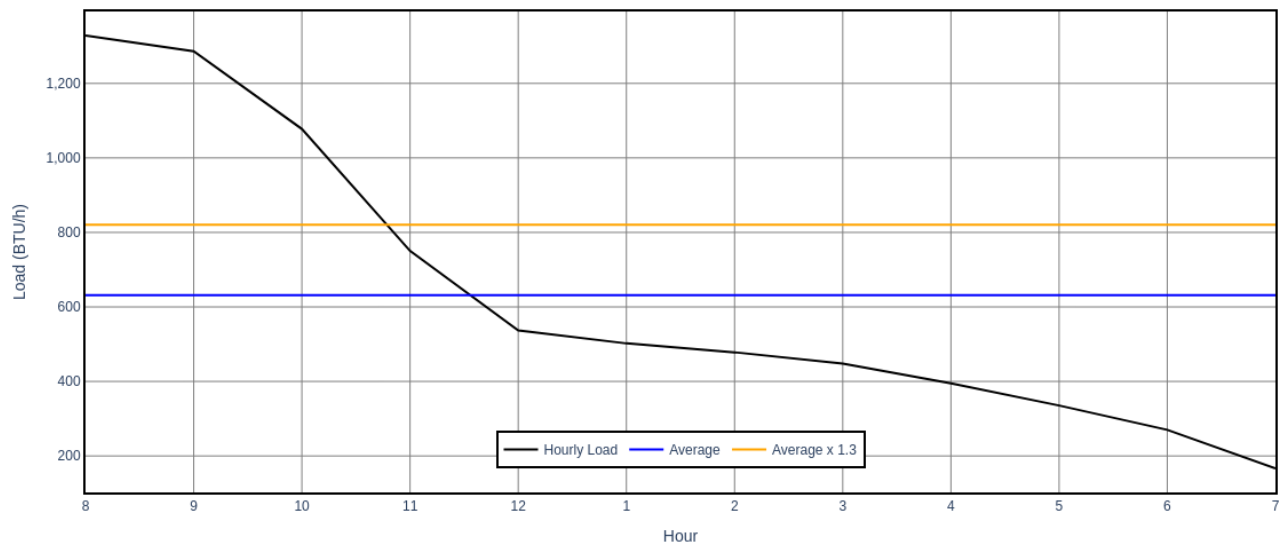
Kitchen Load

First story

• Total Floor Area (ft ²)	235
• Total Volume (ft ³)	1,886
• Ceiling height(s) (ft)	8.0
• Exposed wall area (ft ²) (gross)	226
• Exposed wall area (ft ²) (net)	207
• Total glazing area (ft ²)	19
• Design CFM	
• Occupants	0

	Sensible Cooling BTUH	Latent Cooling BTUH	Total Cooling BTUH	Total Heating BTUH
Total	4,088	585	4,673	5,484
Walls	424	0	424	1,428
Glazing	1,328	0	1,328	751
Doors	0	0	0	0
Ceilings	541	0	541	818
Floors	122	0	122	849
Ventilation	0	0	0	0
Ducts	402	468	871	852
Infiltration Loss and Gain	70	117	187	787
Blower Heat	0	0	0	0
Occupants	0	0	0	0
Appliances	1,200	0	1,200	0
Plants	0	0	0	0
Hot Water Piping	0	0	0	0
Moisture Migration	0	0	0	0
Winter Humidification	0	0	0	0

Kitchen (First) - Hourly Fenestration Load - Midsummer



UNASSIGNED

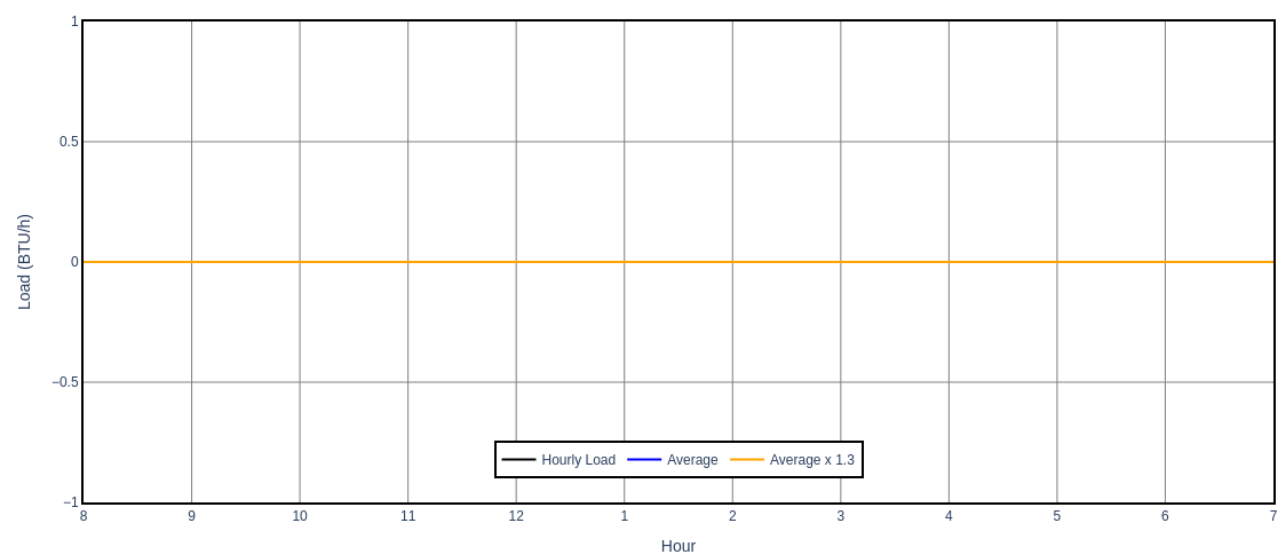
Stairwell To Basement Load

First story

• Total Floor Area (ft ²)	61
• Total Volume (ft ³)	489
• Ceiling height(s) (ft)	8.0
• Exposed wall area (ft ²) (gross)	111
• Exposed wall area (ft ²) (net)	111
• Total glazing area (ft ²)	0
• Design CFM	
• Occupants	0

	Sensible Cooling BTUH	Latent Cooling BTUH	Total Cooling BTUH	Total Heating BTUH
Total	0	0	0	1,581
Walls	0	0	0	763
Glazing	0	0	0	0
Doors	0	0	0	0
Ceilings	0	0	0	212
Floors	0	0	0	220
Ventilation	0	0	0	0
Ducts	0	0	0	0
Infiltration Loss and Gain	0	0	0	386
Blower Heat	0	0	0	0
Occupants	0	0	0	0
Appliances	0	0	0	0
Plants	0	0	0	0
Hot Water Piping	0	0	0	0
Moisture Migration	0	0	0	0
Winter Humidification	0	0	0	0

Stairwell To Basement (First) - Hourly Fenestration Load - Midsummer



UNASSIGNED

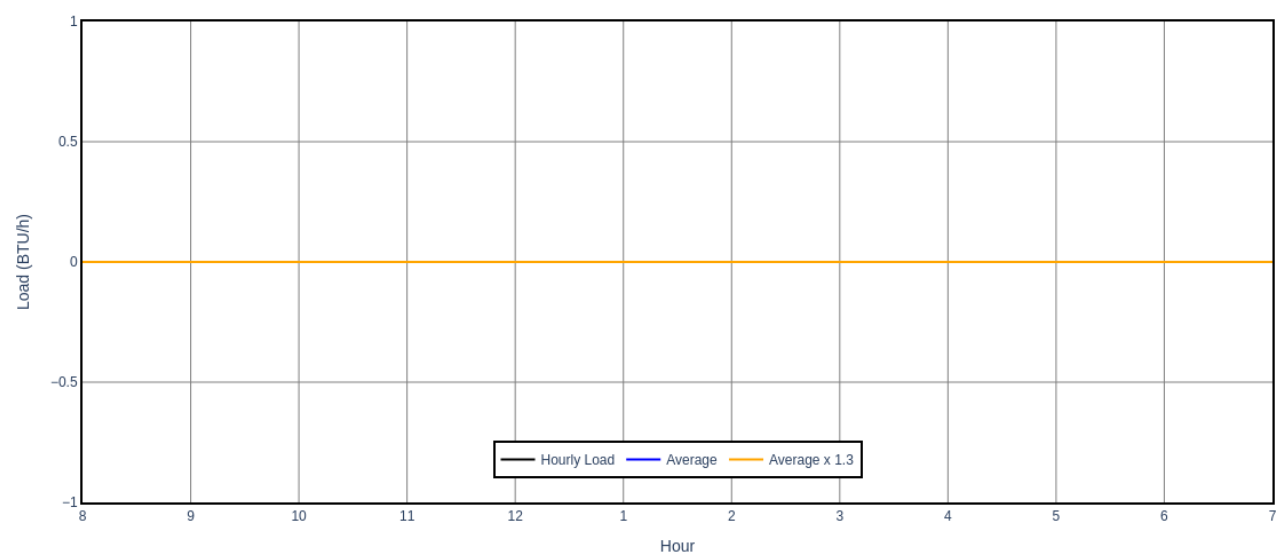
Basement Load

Basement

• Total Floor Area (ft²)	912
• Total Volume (ft³)	7,076
• Ceiling height(s) (ft)	7.8
• Exposed wall area (ft²) (gross)	919
• Exposed wall area (ft²) (net)	919
• Total glazing area (ft²)	0
• Design CFM	
• Occupants	2

	Sensible Cooling BTUH	Latent Cooling BTUH	Total Cooling BTUH	Total Heating BTUH
Total	0	0	0	11,447
Walls	0	0	0	6,693
Glazing	0	0	0	0
Doors	0	0	0	0
Ceilings	0	0	0	0
Floors	0	0	0	1,554
Ventilation	0	0	0	0
Ducts	0	0	0	0
Infiltration Loss and Gain	0	0	0	3,201
Blower Heat	0	0	0	0
Occupants	0	0	0	0
Appliances	0	0	0	0
Plants	0	0	0	0
Hot Water Piping	0	0	0	0
Moisture Migration	0	0	0	0
Winter Humidification	0	0	0	0

Basement (Basement) - Hourly Fenestration Load - Midsummer



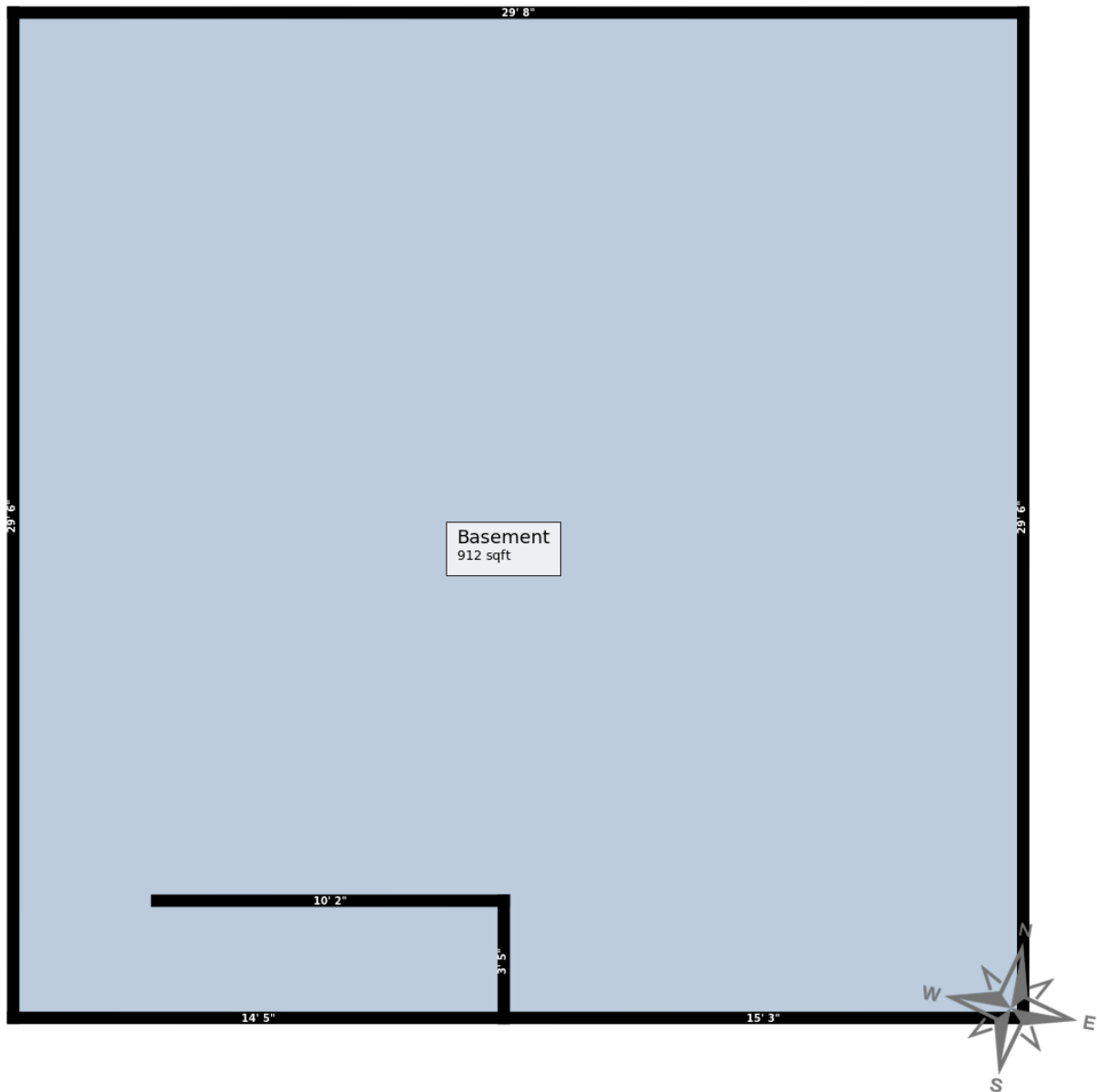
Building Materials Breakdown

Construction Type	Construction Number	Orientation (if applicable)	Area (ft ²)	Cooling BTUH	Heating BTUH	U-Value
Above-Grade Wall	12B-0s w		716	1,466	4,933	0.097
↳ Wall configuration: Exterior, Wall Type: Above-Grade, Primary Structure: Frame Walls, Framing Material: Wood, Frame Construction: 2x4, Cavity Insulation: R-11, Exterior Finish: Stucco or Siding						
Above-Grade Wall	15B0-0-x		0	0	0	0.488
↳ Wall configuration: Exterior, Wall Type: Basement, Primary Structure: 8" Brick, Stone, or Concrete						
Above-Grade Wall	15B0-4sf-x		157	251	1,834	0.165
↳ Wall configuration: Exterior, Wall Type: Basement, Primary Structure: 8" Brick, Stone, or Concrete, Board Insulation: R-4, Insulation Depth: Sill to Floor						
Below-Grade Wall	15B0-0-x		229		1,904	0.117
↳ Wall configuration: Exterior, Wall Type: Basement, Primary Structure: 8" Brick, Stone, or Concrete						
Below-Grade Wall	15B0-4sf-x		533		2,954	0.078
↳ Wall configuration: Exterior, Wall Type: Basement, Primary Structure: 8" Brick, Stone, or Concrete, Board Insulation: R-4, Insulation Depth: Sill to Floor						
Door	11D		0			None
↳ None						
Floor	19A-11p		969	502	3,502	0.073
↳ Floor type: Floor over enclosed unconditioned crawl space or basement, Radiant floor heat: No, Floor insulation: R-11 blanket						
Floor	21A		912	0	1,554	0.024
↳ Floor type: Basement floor, Width of Shortest Side of Floor Slab (Ft): Unknown						
Ceiling	16B-19 ad		969	2,232	3,372	0.049
↳ Ceiling type: Ceiling under attic or attic knee wall, Attic ventilation: FHA vented, Attic fan: No, Radiant barrier: No, Roofing material: Asphalt shingles, Roof color: Dark, R-value: R-19						
Window	1D-c	N	0			None
Window	1D-c	E	34	1,204	1,396	0.57
Window	1D-c	W	0			None

Construction Type	Construction Number	Orientation (if applicable)	Area (ft ²)	Cooling BTUH	Heating BTUH	U-Value
Window	1D-c	S	0			None
Window	1D-c	total	34	1,204	1,396	0.57
↳ Window type: Standard Window, Frame construction: Wood, Wood with Metal Clad or Vinyl, Glass type: Clear, Operability: Operable, Number of panes: Double						

Basement story floor plan

Most common wall height: 7' 9"



First story floor plan

Most common wall height: 8'



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